

Investigation of Delayed Neutron Precursor Data Uncertainties for Microscopic and Summation Calculations

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Abstract

This paper presents an analysis on the sensitivity of [delayed neutron precursor \(DNP\)](#) group parameters, group half-lives τ_k and yields $\nu_{d,k}$, and summed delayed neutron yields to uncertainties in their nuclear data for various datasets, including the [International Atomic Energy Agency \(IAEA\)](#) beta-delayed neutron emission database, ENDF/B-VII.1, ENDF/B-VIII.0, ENDF/B-VIII.1, and JEFF-3.1.1. These uncertainty-weighted sensitivities provide a detailed view of which [DNPs](#) need more accurate data such that analyses using individual [DNPs](#), such as microscopic group parameter formulation and summation calculations, may be more accurate. Improving the accuracy of these analyses is of particular importance for kinetics models, which rely on accurate [DNP](#) group parameters. The microscopic approach for group parameter calculation is useful for providing insight into how each [DNP](#) contributes to delayed neutron emission, but it requires accurate nuclear data. The analyses conducted in this work reveal several [DNPs](#) that correlate strongly with the [DNP](#) group parameters, indicating a need for more accurate data for those [DNPs](#). There are also [DNPs](#) which have large uncertainties and vary significantly between datasets which also warrant additional attention. This paper informs which [DNPs](#) are in need of updates to their data for various nuclear datasets.

Keywords — Delayed Neutron Precursors, Group Parameters, Nuclear Data, Nuclear Data Uncertainties, ENDF, JEFF

Is it worth talking about MoS-DeN here? Maybe suggest that this is useful for advanced reactors? It might be useful for motivation.

I. INTRODUCTION

- Accurately modeling nuclear reactor kinetics is important for safe operation and understanding reactor responses to various transients
- DNPs are a key component of kinetics modeling (say what they are, typically use 6 groups, introduce notation and group parameters, microscopic calculations as an approach to calculate these parameters)
- However, there is a need for more data on DNPs.
- This paper aims to clarify and specify what DNP data are lacking in different datasets, offering weighted values that indicate not only the uncertainty of the data, but the relative impact of that uncertainty on the calculations that use it.
- (Tie it back into the broad scope) By providing the data of importance for these datasets, future analyses can target experiments to produce that specific data to further improve DNP group parameters and kinetics models.

II. METHODS

Overview - I calculated the DNP group parameters using the microscopic approach similar to how it is conducted in the literature (give references) using an in-development Python package called MoSDeN (maybe don't discuss MoSDeN since this isn't molten salt related?). However, instead of using a pulse irradiation approach, I use CFYs and such (faster, no depletion chain; enabled running many comparisons and samples to do sensitivity studies; easier to quantify uncertainty using CFY rather than propagating through depletion chain). I then used that sampled data to calculate the delayed neutron count rate and used that delayed neutron count rate to calculate the group parameters. I repeated this approach 5000 times for each dataset to calculate the sensitivity of the group parameters to variation of the DNPs within their uncertainties, assuming a normal distribution about the standard deviation.

II.A. Data

- Explain the CFY, P_n , and λ values.

- Discuss the various datasets that will be included (combinations of datasets only considered for datasets that do not include certain data (such as IAEA database not including CFYs))

II.B. Solver

- Explain delayed neutron count rate calculation (from data and from groups) (with sampled params)
- Discuss that the microscopic approach enables evaluation of specific DNPs and their impacts (rather than using a macroscopic approach, which only uses the count rate)
- Discuss how this approach enables capturing the sensitivities of individual DNPs and isolating problematic data
- Discuss that this approach (compared to using the OpenMC approach) is faster, doesn't require depletion chain modeling, and enables a per-nuclide evaluation of what data needs investigation. It also pairs well with summation calculations, as it uses the same data.
- Explain non-linear least squares solve
- Discuss other ways in literature (known nuclides, exponential stripping) and justify that this approach offers a less subjective methodology, though the use of known nuclides could also be improved with additional data on other DNPs

II.C. Sensitivity Analyses

- Discuss Pearson Correlation Coefficient usage (give an example figure for -1, 0, and +1) and uncertainty weighted calculation
- PCC was selected to determine variation in group parameters due to variation in individual DNP data
- This approach was selected because the sensitivities did not generally have strong non-linear behavior. The non-linear least squares will result in non-linear effects, but linear correlations dominate in sensitivities.

III. RESULTS

Each sub-section in the results will read pretty-similarly.

- Show the largest PCCs in a table
- Show chart of nuclides with summed PCCs and summed uncertainty-weighted PCCs
- Show bar chart with most significant summed uncertainty-weighted PCCs
- Discuss which data contribute most to any outliers, and what DNPs need data more than others.

III.A. IAEA Beta-Delayed Neutron Emission Database

- Show that results using CFYs from JEFF-3.1.1 match results from IAEA CRP document (verification with literature)
- Discuss impacts from other CFYs (potential issues with specific DNP CFYs)

III.B. ENDF/B-VII.1

III.C. JEFF-3.1.1

IV. CONCLUSIONS

- Summarize the main takeaways from the results (maybe a table indicating DNPs of interest for different datasets?)
- Describe how improving these data will enable development of improved models of DNPs and kinetics models

Possibly tie
into MSRs
and MoS-
DeN?

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